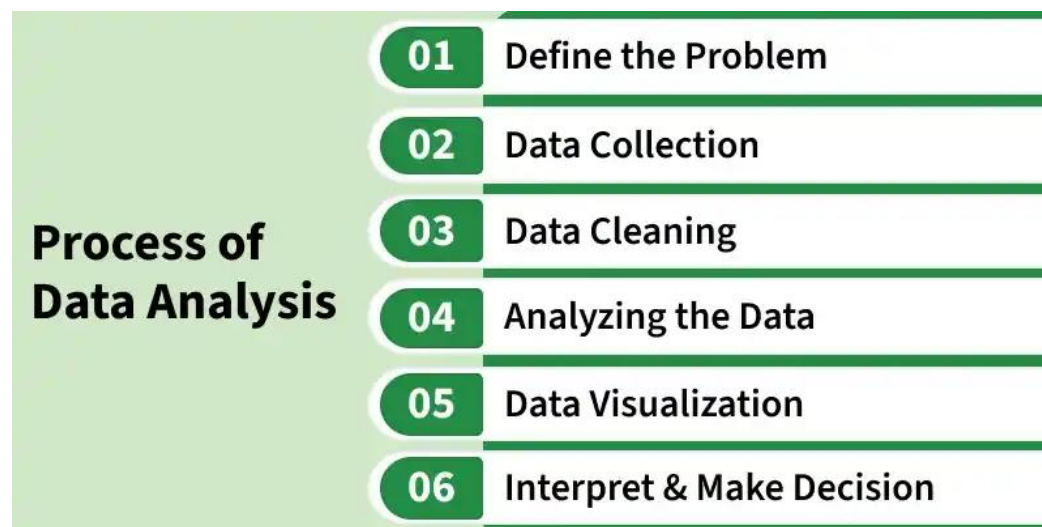


DATA ANALYTICS

Data analytics is process of examining the raw data to uncover pattern and draw a meaningful insight and support better decision making. It helps individuals and businesses understand past performance monitor current trend and predict future outcomes and it will help to identify trends patterns of the data to make informed decisions

STEPS



1.DEFINE THE PROBLEM

Before starting any analysis, it's crucial to clearly understand what we are trying to solve. This involves defining the specific question, goal or problem and aligning it with stakeholder's expectations. A well-defined problem ensures that the analysis stays focused and relevant.

2.DATA COLLECTION

Once the problem is defined, the next step is to gather data from relevant sources. This may include internal databases, surveys, web scraping or publicly available datasets like Kaggle. Collecting the right data ensures that the analysis is grounded in accurate and comprehensive information.

3. DATA CLEANING

Raw data is rarely ready for analysis. This step involves handling missing values, removing duplicates, standardizing formats and converting categorical variables into numerical forms. Properly prepared data improves the reliability and accuracy of our insights.

4. ANALYSE THE DATA

Data analysis is the core step where patterns, trends and relationships are uncovered. Depending on the objective, this may involve descriptive statistics, correlation analysis or advanced machine learning models to extract meaningful insights

5. VISUALIZE THE DATA

Visualizations make complex data understandable. By using charts, graphs and dashboards, we can highlight key insights, patterns and outliers. Effective visualization simplifies decision-making and communication to stakeholders.

6. INTREPRET AND DECISION MAKING

The final step is translating analytical results into actionable insights. Interpretation involves contextualizing findings, communicating them effectively and making data-driven decisions. Continuous monitoring ensures that the decisions remain effective over time

DATA ANALYTICS TOOLS

Data analytics has rapidly evolved in technological capabilities, in addition to a broad range of mathematical and statistical approaches to crunching numbers. Data analysts have a broad range of software tools to help acquire data, store information, process data, and report finding

1.PYTHON

Python is a versatile and often favoured programming language for data analysis and scientific computing. With a rich ecosystem of libraries like pandas, NumPy, and Matplotlib, Python facilitates complex data manipulation, analysis, and visualization with relative ease.

2.MICROSOFT EXCEL

Microsoft Excel, fundamentally spreadsheet software, also has noteworthy data analytics capabilities. Because of the wide enterprise-level adoption of Microsoft products, many businesses already have access to it.

3. TABLEAU

Tableau stands out as a leading data visualisation software widely utilised in business analytics and intelligence. Tableau is a popular data visualisation tool for its easy-to-use interface and powerful capabilities. Its software can connect with hundreds of data sources and manipulate the information in many different visualisation types. It uniquely appeals to business users, who appreciate its simplicity and centralised platform. Data analysts also appreciate more advanced big data tools for tasks such as clustering and regression.

4.SQL

SQL (Structured Query Language) is the primary language used to communicate with databases and is considered a "must-have" skill for data analysts. It allows users to perform tasks that standard spreadsheet software often cannot handle, such as managing millions of rows or connecting complex data from multiple tables.

5. GOOGLE ANALYTICS

Google Analytics is a tool that helps businesses understand how people interact with their websites and apps. You add a special JavaScript code to your web pages to use it. This code collects information when someone visits your website, like which pages they see, what device they're using, and how they found your site. It then sends this data to Google Analytics, which organises it into reports. These reports help you see patterns, like which products are most popular or which ads bring people to your site.

6.GOOGLE CHARTS

Google Charts is a free online tool that excels at producing an array of interactive and engaging data visualisations. Its design caters to user-friendliness, offering a comprehensive selection of pre-set chart types that can be embedded into web pages or applications.